

- 7 Alpha-1 antitrypsin (AAT) deficiency is an inherited disease that can affect different organ systems.

Alpha-1 antitrypsin (AAT) is a protein produced in the liver.

(a) Investigation of the breathing of a patient with AAT deficiency found that the:

- resting tidal volume (V_T) was 500 cm^3
- dead space (DS) was 150 cm^3
- resting respiratory rate (RR) was 12 breaths per minute.

Calculate the volume of air that enters the alveoli (A_v) **each hour**.

$$A_v = (V_T - DS) \times RR$$

Give your answer to 2 significant figures with appropriate units.

(3)

Answer

(b) Activity of the heart was investigated in the same patient.

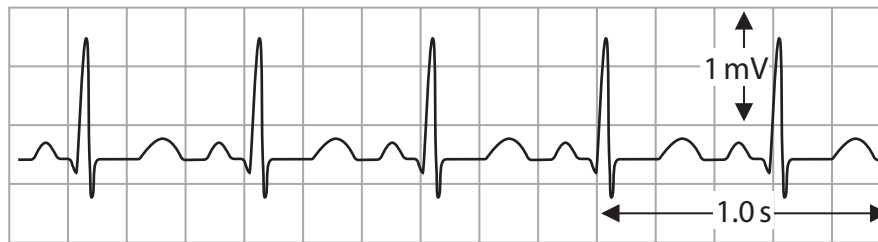
The trace shows a normal electrocardiogram (ECG).



(i) Name the stage of the cardiac cycle at T on this ECG.

(1)

(ii) The ECG of the patient with AAT is shown in the trace.



Calculate the mean heart rate shown in this ECG trace.

(1)

Answer

(iii) Compare and contrast the ECG trace of this patient with the normal ECG trace.

(3)

[illegible]

- (c) One potential treatment for AAT deficiency uses adenoviruses to genetically modify cells in an affected person.

Adenoviruses contain double-stranded DNA.

Explain how adenoviruses can be used to treat AAT deficiency in a person.

(4)

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(Total for Question 7 = 12 marks)